

DETAILED REPORT

V-GUARD INDUSTRIES LTD - BIG IDEA TECH DESIGN COMPETITION 2026

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Synopsis

India's homes still run on unreliable power, and the home inverter with its battery is the lifeline that keeps lights, refrigeration, connectivity and medical devices alive. Yet even today's Wi-Fi enabled inverters remain reactive: they report the present state but cannot predict failure, understand load, or decide for themselves. Batteries therefore die without warning, often mid-blackout; backup drains on non-essential loads; and warranty is a dispute rather than a service. V-Guard Sentinel closes this gap by embedding a low-cost, on-device AI Core inside the inverter that predicts its battery's gradual ageing weeks ahead as a graded warning (and flags sudden faults), manages charging and the essential circuit use, and detects grid disturbances - all offline, with no dependence on the internet. The models sharpen as V-Guard's own fleet data grows - a proprietary-data moat competitors cannot rebuild - with optional privacy-preserving federated learning for connected units. This report elaborates the Phase-2 concept into an engineering-grade design: the physics that makes battery degradation measurable and therefore learnable; a system and hardware architecture built around galvanic isolation and proven TinyML silicon; the machine-learning method stack for state-of-health and remaining-useful-life estimation, with an honest account of the lead-acid data gap and prediction limits; the load-management logic; and a deployable learning pipeline. It defines the electrical, computational, model and product parameters to be specified, a software-led prototype and validation plan, the applicable Indian and international standards, a candid incremental-versus-standalone cost analysis, and the market, sustainability and business case. Throughout, the design is deliberately scoped to the boundary between what is deployable today and what remains a research target, because credibility, not hype, is what turns an intelligent product into a trusted one. Sentinel reframes V-Guard from the company that protects India's power into the intelligence layer the Indian home runs on.

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1. Introduction

The Big Idea Tech 2026 theme asks how V-Guard moves from smart products to intelligent products. Our Phase-2 submission proposed V-Guard Intelligence: a low-cost, on-device AI Core that lets each V-Guard product predict its own failures, learn its owner's habits, and act autonomously, with the whole installed base improving together through privacy-preserving federated learning, all working offline. The flagship is V-Guard Sentinel, an intelligence upgrade to the home inverter and its battery. This detailed report is a faithful continuation of that proposal - the concept is unchanged - elaborated into an engineering-grade design. Where this report states a capability or number more precisely than the executive summary - detection rather than prediction of grid sags, backup gains given as a range, cost split into incremental and standalone - these are refinements within the same concept for technical accuracy, not changes of scope. It covers the problem physics, the system and hardware architecture, the machine-learning methods and their honest limits, the parameters to be defined, a prototype and validation plan, standards and safety, unit economics, and the rollout. A deliberate thread runs through it: we separate what is deployable today from what is a research target, because the report is read by engineers, and credibility is the whole game.

2. Problem Context and Motivation

2.1 India's power-reliability gap

Independent surveys place the lived experience of Indian households far below official feeder averages. An estimated 85% of households face outages daily and 37% experience two to eight hours of outage per day [1]; one in three households suffers a blackout, low-voltage or appliance-damage event within any single month [2]. Government figures showing roughly 22.6 hours per day of rural supply conceal the evening collapse, when rural supply between 5 and 11 PM averages only about 4.7 hours [3]. Household voltage routinely swings between 180 V and 270 V against a 230 V nominal. For tens of millions of homes the inverter and battery are therefore not a convenience but critical infrastructure.

2.2 The battery as the silent point of failure

The tubular lead-acid battery, still around half of the Indian inverter-battery market, is the component that strands users. Most of its failure is gradual and measurable - the months-long end-point of irreversible sulphation from chronic undercharge, positive-grid corrosion, water loss and dry-out, active-material shedding, and acid stratification [4] - and only a minority of modes, such as a corrosion-driven open circuit, strike with little warning. Lithium-ion batteries degrade by different mechanisms - solid-electrolyte-interphase growth, and lithium plating under cold, fast

charge - but the principle holds: degradation is progressive and observable [5]. The single most important stressor in India is temperature: lead-acid life roughly halves for every 10 degrees Celsius above 25 [6], so a battery rated for ten years at 25 C ages about three times faster at 41 C (~3.3 years), and field life is shorter still under real duty. This is precisely the environment V-Guard sells into, and precisely why proactive health management matters here more than anywhere.

2.3 Why degradation is predictable, and why today's products miss it

Each ageing mechanism leaves an electrical fingerprint an inverter can observe: rising internal resistance, worsening voltage sag under load, declining charge acceptance and coulombic efficiency, and shifts in the rested open-circuit-voltage curve. Internal-resistance rise is the primary online state-of-health proxy, and charge-acceptance decline is a cheap, high-information lead-acid signal [4][6]. No single signal uniquely identifies a mechanism, but their combination tracks health closely (Exhibit 5). Today's inverters, even Wi-Fi models, measure at best the present state; none forecast the trajectory. So the four unmet needs are foresight of failure, load intelligence during outages, energy insight as time-of-use tariffs arrive, and proactive rather than reactive service. V-Guard, founded in 1977 to solve exactly Indian power unreliability, manufactures its own batteries, has taken a 30.35% stake in a battery-technology company, and has opened a 120-crore Kochi innovation campus [25]; no competitor is better placed to close this gap.

2.4 The cost of the status quo

The status quo is quietly expensive: a battery that fails early costs the household a replacement and, inside the flat-warranty window, costs V-Guard directly; a blackout that outlasts a mismanaged battery spoils food and interrupts work, study and medical routines; and voltage excursions shorten every appliance's life. Because today's inverters never turn the data they already sense into foresight, this loss is simply absorbed. Intelligence embedded in the product, not bolted on through an app, turns that sunk cost into avoided cost.

3. Proposed Solution: V-Guard Sentinel

Sentinel is an intelligence layer, not a new gadget. A compact AI Core - a microcontroller with a neural accelerator plus a small sensor set - is embedded in new inverters and offered as a retrofit module for the installed base. It runs four on-device engines and participates in a privacy-preserving federated-learning loop (Exhibit 1). Engine 1, Predictive Health and Longevity, estimates true battery state-of-health and predicts a remaining-useful-life window, flags degrading internals early, and applies an adaptive charge-and-thermal policy to extend life; a signed, append-

only, device-key-authenticated health-log turns warranty into a data-backed, one-tap process. Engine 2, the Habit-Learning Autopilot, learns the home's load and outage patterns and, during a blackout, autonomously protects critical loads while shedding the rest. Engine 3, Energy Coach and Grid Shield, discloses where power goes and detects grid disturbances for appliance protection. Engine 4, Fleet Intelligence, sharpens the shared models as V-Guard's proprietary fleet data grows - centrally, with optional privacy-preserving federated learning for connected units. The same compute platform - each product with its own sensors and model - extends across pumps, stabilisers and water heaters, turning a portfolio of smart products into one intelligent, connected home.

4. System Architecture

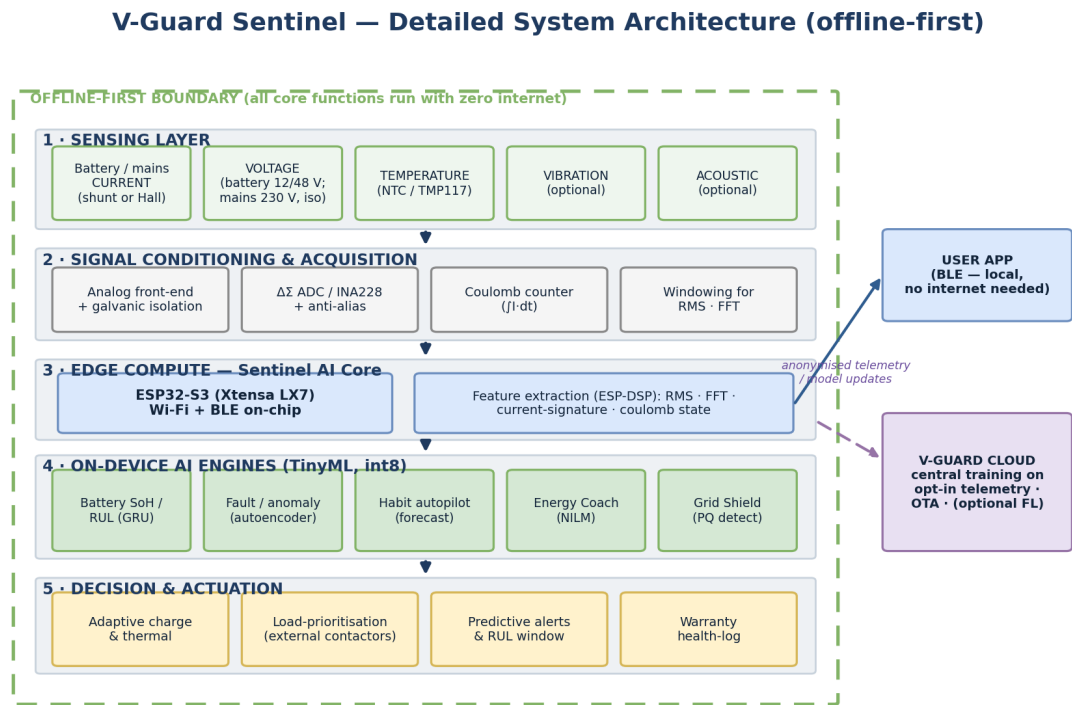


Exhibit 1. V-Guard Sentinel detailed system architecture: existing inverter sensors feed on-device TinyML engines that drive autonomous actions; only anonymised telemetry or model updates leave the home, and all core functions run offline.

Sentinel is organised as five on-device layers inside an offline-first boundary, with opportunistic links to a phone app over Bluetooth Low Energy and to the V-Guard cloud over Wi-Fi. The sensing layer captures current (via an external shunt or Hall sensor), isolated voltage and temperature, with optional vibration/acoustic sensing for inverter-side fault detection. The signal-conditioning layer provides the analog front-end, galvanic isolation, a high-resolution ADC, anti-alias filtering, a coulomb counter, and windowed buffers for FFT and statistics. The edge-compute

layer is the AI Core itself. The AI-engine layer hosts the quantised TinyML models, and the decision-and-actuation layer performs adaptive charging, relay-based load prioritisation, alerting and health-logging. Every core function runs with zero internet; the cloud is a bonus for federated aggregation, over-the-air updates and analytics. This ordering - sensing, conditioning, compute, inference, action - mirrors a conventional embedded-control stack, which keeps the design conventional to manufacture while the intelligence sits in software.

5. Hardware Design of the AI Core

V-Guard Sentinel AI Core — Hardware Block Diagram

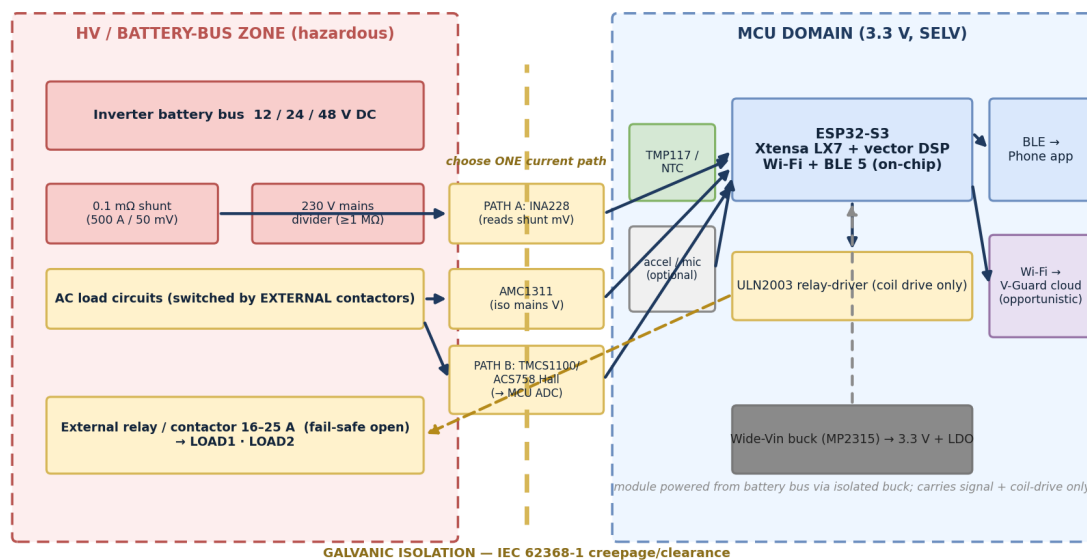


Exhibit 2. AI Core hardware block diagram. A galvanic-isolation barrier separates the hazardous battery-bus/HV zone from the SELV MCU domain; isolation ICs (INA228, AMC1311, TMCS1100) bridge it safely.

5.1 Compute

The workloads - a small recurrent or 1-D convolutional network for RUL, an autoencoder for anomalies, FFT-based power-quality analysis, and light load-monitoring - fit comfortably within TinyML budgets of tens of kilobytes at int8 precision. The cost-optimal choice is the Espressif ESP32-S3: dual vector-capable cores with integrated Wi-Fi and Bluetooth Low Energy on one part near 300 rupees, which eliminates a separate radio and its 200-to-400-rupee bill-of-materials line [30]. The same architecture drops onto a higher-tier MCU if a future variant ever needs more compute, with no other change. Feature extraction uses Espressif's ESP-DSP library, where a 1024-point fixed-point FFT runs in about one millisecond on the LX7 vector core [12]. Event detection (sag/swell) uses a one-cycle time-domain RMS; the multi-cycle FFT is reserved for harmonics, so the two run at different windows without conflict.

5.2 Isolated sensing front-end

Safety is the backbone of the design. The battery bus is low-voltage DC (12/24/48 V), but anything touching the AC mains - sensed across a 150-to-300-volt range - must be galvanically isolated from the microcontroller domain; and, critically, the 50-to-150-ampere battery current must never flow through the module's small PCB. Following the architecture of every commercial battery monitor - a separate high-current shunt or Hall sensor feeding a signal-only electronics head, exactly as in the Victron SmartShunt whose head (69 by 69 by 31 millimetres) is almost identical to our enclosure [32] - the Core reads current only as a low-level signal, via one of two mutually-exclusive paths: either a shunt (the inverter's existing one, or an external 500-ampere / 50-millivolt, about 0.1-milliohm, busbar shunt) read by a 20-bit charge-accumulating monitor (INA228; the 16-bit INA226 is the cost-tier option) that sees only tens of millivolts, or a contactless ACS758/TMCS1100 Hall sensor whose ratiometric output goes straight to the MCU ADC [30]. An ordinary split-core current transformer will not work, since transformers sense only alternating current; the direct-current battery line needs a busbar shunt or a Hall sensor. Bus voltage is sensed through a high-value divider into a reinforced isolation amplifier (AMC1311) via thin leads fused at one ampere at the battery terminal for safety. Temperature uses a battery-terminal NTC thermistor plus an optional 0.1-degree digital sensor (TMP117). Battery health needs no vibration or acoustic sensing; an optional accelerometer or MEMS microphone is offered only as an industrial add-on for inverter-side fan, relay or transformer fault detection, not part of the battery-health core or the default bill of materials. For coulomb counting the binding constraint is not ADC resolution but current-sense offset, so the design relies on the INA228's built-in offset calibration and re-anchors state-of-charge against open-circuit voltage during the float-rest periods between outages.

5.3 Actuation, power and mechanical

Load shedding drives external electromechanical relays or contactors rated 16 to 25 amps - chosen for low cost and a fail-safe galvanic open - via a ULN2003 array and isolated coils, so the module carries only the coil-drive signal, not the switched current; solid-state relays are reserved for cases needing fast, frequent switching. A wide-input synchronous buck converter (MP2315-class) taps the inverter's existing rail to produce a 3.3-volt supply with a microamp-class quiescent budget, so the module never meaningfully loads the battery. The mechanical package is a flame-retardant PC/ABS enclosure rated UL94 V-0, roughly 90 by 70 by 35 millimetres - the extra height over a minimal board giving relay and connector clearance - housing an 80-by-60-millimetre PCB with four M3 mounting points, indoor IP20 rating, DIN-rail or adhesive mounting, and segregated screw terminals that carry only signal and relay-drive lines (not main current) plus JST connectors for low-voltage leads (Exhibit 3). The module mounts on or beside the inverter chassis, which has

ample room in its roughly 275-by-250-by-120-millimetre envelope, and never on the 40-to-65-kilogram floor-standing battery; high-current load circuits are switched by external contactors driven from the module, so that current too stays off the PCB.

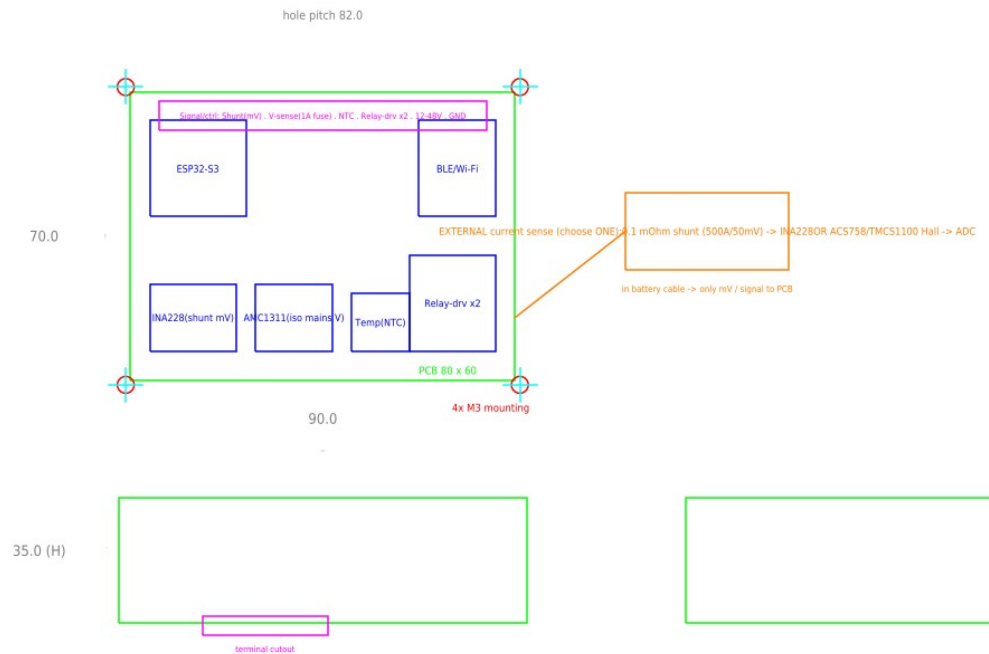


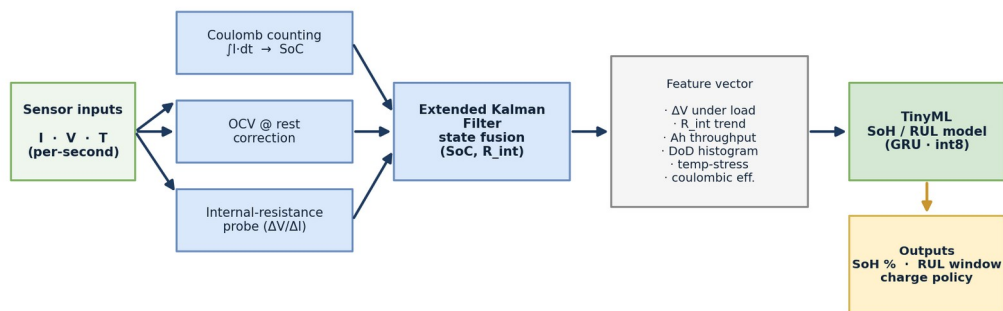
Exhibit 3. Sentinel AI Core - early-stage dimensioned layout (top/front/side views). Editable AutoCAD sources (DXF, parametric AutoLISP, script) are in the CAD folder.

5.4 Cost: the incremental-versus-standalone reality

An itemised bill of materials (Exhibit 15) shows the basic board at about 500 rupees, or roughly 800 rupees standalone once an external current shunt is added where the inverter lacks one; a premium board with precision isolated sensing and a higher-current contactor stage is roughly two to three times that. The headline target of 150 to 800 rupees of added cost is realistic as an incremental cost over an inverter that already carries a shunt, relays, a microcontroller and a power rail - which metering-capable inverters do - because those parts are then shared. Stating this openly is deliberate: an over-optimistic single number would not survive expert scrutiny, whereas the incremental-cost argument is both true and strategically stronger.

6. Engine 1: Predictive Battery Health and Remaining Useful Life

Battery State-of-Health & Remaining-Useful-Life Estimation — Signal Flow



trained on NASA / CALCE + refined via federated fleet learning

Exhibit 4. Battery SoH/RUL signal flow: coulomb counting, OCV correction and an internal-resistance probe are fused by an Extended Kalman Filter into a feature vector consumed by a compact TinyML model.

6.1 Estimation stack

Sentinel fuses classical estimation with a small learned corrector. Coulomb counting integrates current for a fast state-of-charge estimate but drifts because it integrates sensor bias open-loop; a 0.5% current bias can produce several percent of drift per week. Rested open-circuit voltage and an internal-resistance estimate - from the delta-V/delta-I of natural load and charge transients (or a cooperative pulse from the host charger), temperature- and state-of-charge-normalised before use as a health signal - correct it, and an Extended Kalman Filter fuses these into a bounded state, an approach that is mainstream in production battery management and reports state-of-charge RMSE around one percent [8][9] (Exhibit 6). Lead-acid has a usefully sloped open-circuit curve, so voltage anchoring works well; lithium-iron-phosphate's flat plateau makes it harder, an asymmetry the design accounts for. A feature vector - voltage sag under load, internal-resistance trend, ampere-hour throughput, depth-of-discharge histogram, temperature-stress integral and coulombic efficiency, plus incremental-capacity peaks from the partial charge that follows each outage - feeds a compact quantised model that maps feature history to state-of-health and an RUL window.

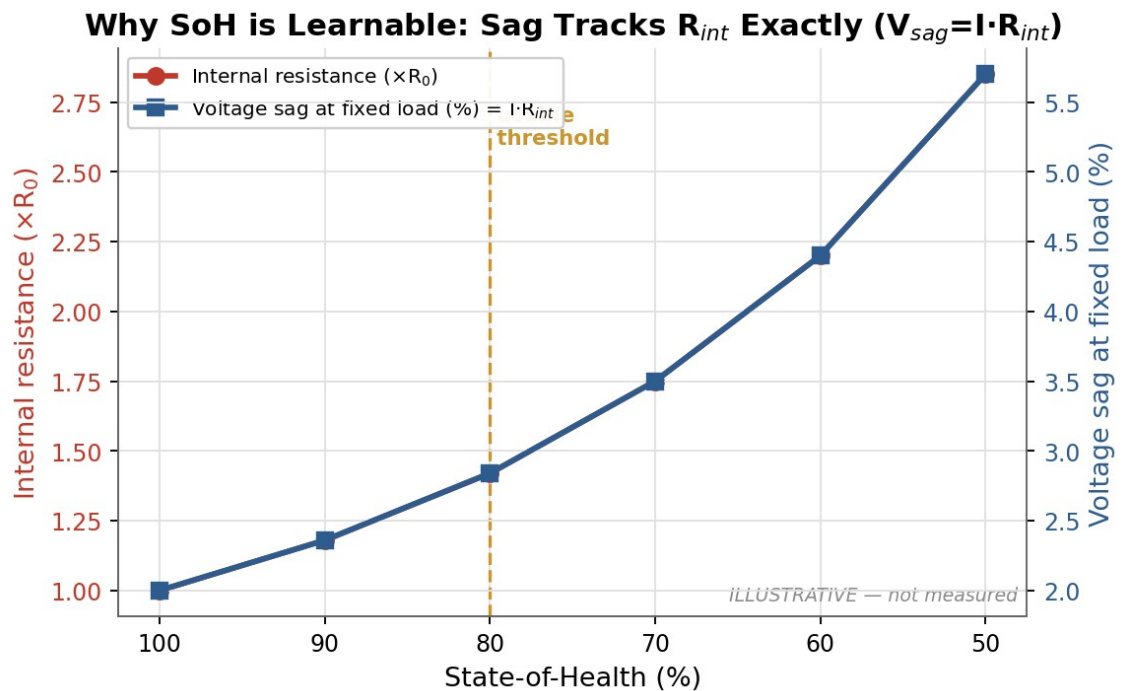


Exhibit 5. Why state-of-health is learnable: internal resistance and voltage-sag-under-load both track degradation measurably as capacity falls toward the 80% usable threshold.

State Estimation: EKF Bounds Coulomb-Counting Drift (simulation)

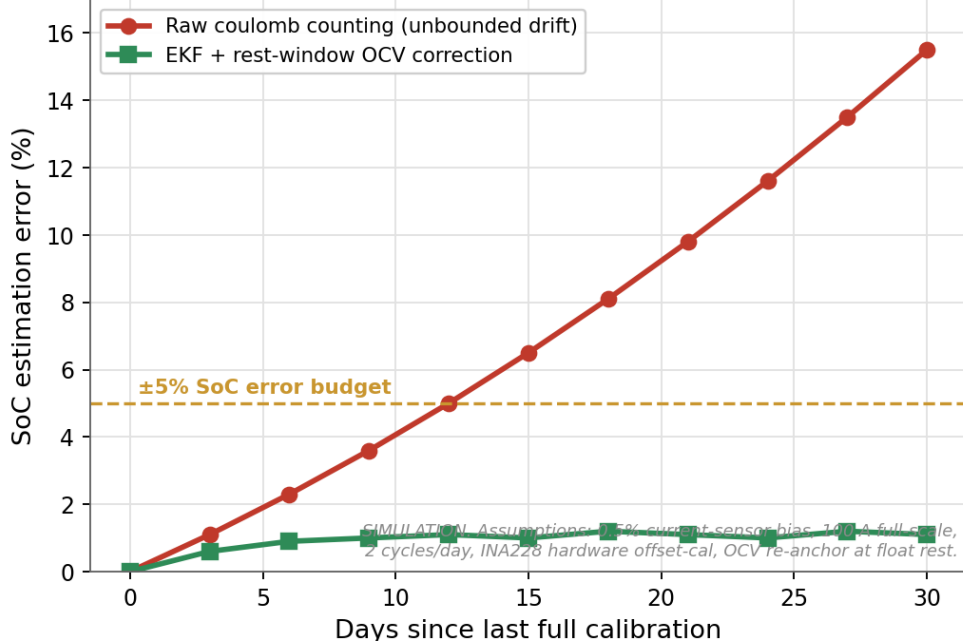


Exhibit 6. State estimation: an Extended Kalman Filter bounds the open-loop drift of raw coulomb counting within a $\pm 5\%$ state-of-charge budget by fusing open-circuit-voltage and internal-resistance corrections.

6.2 Data strategy and the lead-acid gap

We are candid about a real limitation: rigorous, benchmarked machine-learning results for battery health are almost all on lithium-ion laboratory cells, drawing on public datasets such as NASA's

Prognostics Center of Excellence set, CALCE, Oxford and Sandia [7][8][9][10]; there is essentially no canonical open lead-acid degradation dataset. For a tubular-battery product we will therefore run our own accelerated aging campaign - logging partial charge curves, DC internal resistance, charge acceptance, temperature, throughput and depth-of-discharge histograms - and use the abundant lithium-ion data for architecture prototyping and transfer learning rather than direct parameter reuse. V-Guard's in-house battery manufacturing and reliability lab make this aging campaign uniquely practical, and it is itself a defensible moat: the data does not exist to be bought.

The aging campaign is a concrete, ownable programme rather than a hope. A representative matrix of tubular cells is cycled across the temperature, depth-of-discharge and charge-rate combinations that characterise Indian use, instrumented for partial-charge curves, DC internal resistance and charge acceptance, until each reaches the eighty-percent end-of-life threshold. Lithium-ion public datasets seed the model architecture and feature engineering, and transfer learning adapts the pretrained network to lead-acid with far less data than training from scratch would need. Because V-Guard manufactures the cells and runs a reliability lab, this data - the true scarce asset in battery prognostics - is generated in-house and continually enriched by the fleet, compounding the moat described in Section 9.

6.3 On-device feasibility

On-microcontroller inference is well within reach. TensorFlow Lite Micro needs only tens of kilobytes [11], and a published battery-RUL model has run in 11 kilobytes of flash and 1.2 kilobytes of RAM with two-millisecond inference on a sub-dollar Cortex-M0-plus core [13]; the vendor's optimised ESP-NN kernels give a further several-fold speed and energy gain on the LX7 vector core [12]. Because health updates once per charge or per hour, latency is irrelevant; the binding constraints are sensing quality and training data, not compute. Reported state-of-health error is roughly one-to-two percent on clean lithium-ion benchmarks and around five percent on-device after quantisation, and coarser for data-limited lead-acid - numbers we quote as ranges, not as marketing absolutes.

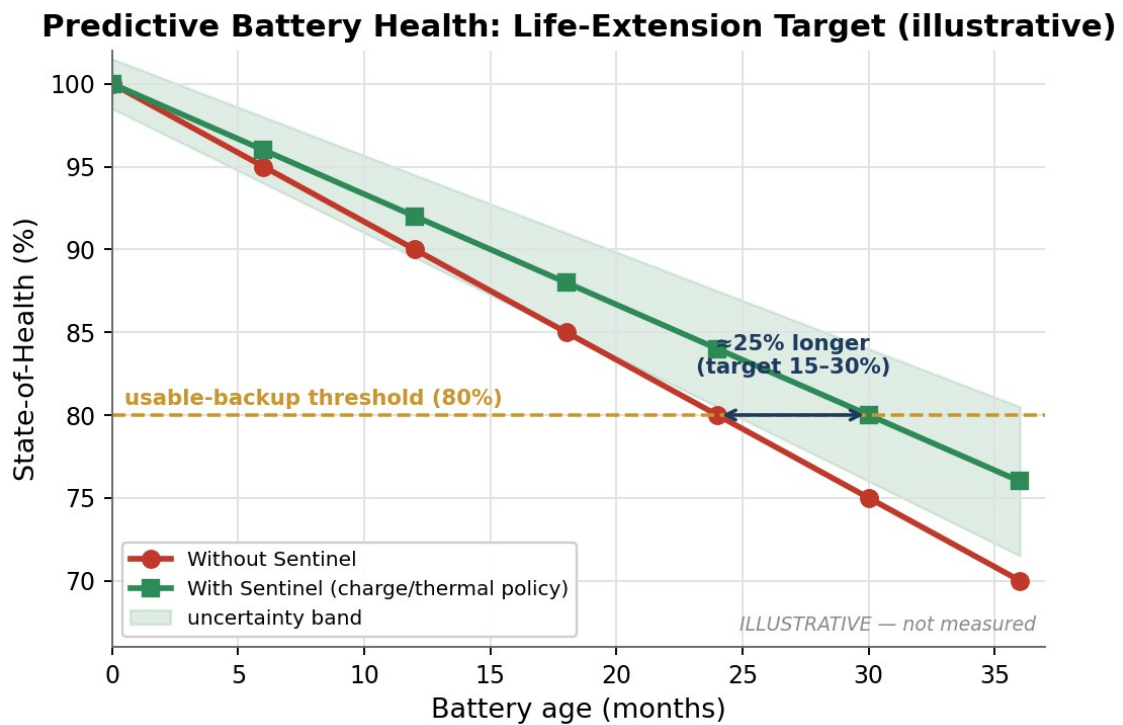


Exhibit 7. Predictive battery health: Sentinel's charge and thermal policy keeps state-of-health above the usable threshold longer, and forecasts the replacement window weeks ahead.

6.4 Honest lead-time framing

The strongest published evidence, on lithium-iron-phosphate cells under controlled cycling, predicts total life from the first hundred cycles at about nine percent error [7]. But three hard limits must be stated. The capacity fade 'knee' is a fundamental barrier, and nominally identical cells diverge widely. RUL is path-dependent on future usage that cannot be seen in advance, so irregular Indian inverter duty will be noisier than the lab. And certain faults - internal shorts, corrosion open-circuits, thermal runaway - are event-driven and not predictable from voltage, current and temperature trends. Sentinel therefore advertises a degradation-trend warning, not a safety-event prediction: it reports state-of-health with an uncertainty band, issues a graded early warning - healthy, degrading, or replace within N weeks - months ahead for gradual wear, and raises anomaly flags for fast faults. This honesty is a feature, because a warning window the installer can trust beats a false-precision date that discredits the product on its first miss.

7. Engine 2: Habit-Learning Autopilot and Load Prioritisation

Load prioritisation is mature but, tellingly, everything shipped commercially - from Tesla Powerwall's essential-circuit logic to smart-panel products - is rule-based on state-of-charge, voltage or frequency thresholds; fully learned, reinforcement-driven control is still a research result with simulation-inflated numbers, and it is entirely absent from Indian home inverters [31].

Sentinel therefore ships a defensible hybrid: robust rule-based tiers made smarter by a learned demand-and-outage forecaster. Its first value is universal and needs no rewiring: it pre-charges the battery ahead of a predicted outage, manages charging to protect battery health, and gracefully manages the single essential sub-circuit that a typical Indian inverter already backs up - a few lights, fans and often the refrigerator - warning the user and prioritising within it as charge runs low. A caveat we state plainly: in most Indian homes the air-conditioner and geyser are not on the inverter at all, so there is little to shed on those tiers. Where a home does carry non-critical load on the inverter - larger systems, or new inverters where V-Guard designs a multi-circuit output - the same logic sheds and defers through external contactors: keep (refrigerator, lights, Wi-Fi, medical), defer (geyser, pump), shed (air-conditioner), re-evaluating until the grid restores, then recharging optimally (Exhibit 8). Where such sheddable load exists this extends usable backup for essentials by of the order of ten to twenty percent - a conditional, not universal, gain - never the sixty-percent figures of simulation-only papers. Fail-safe defaults ensure a sensor or model fault never strands critical loads.

Autonomous Blackout Load-Prioritisation — Control Logic

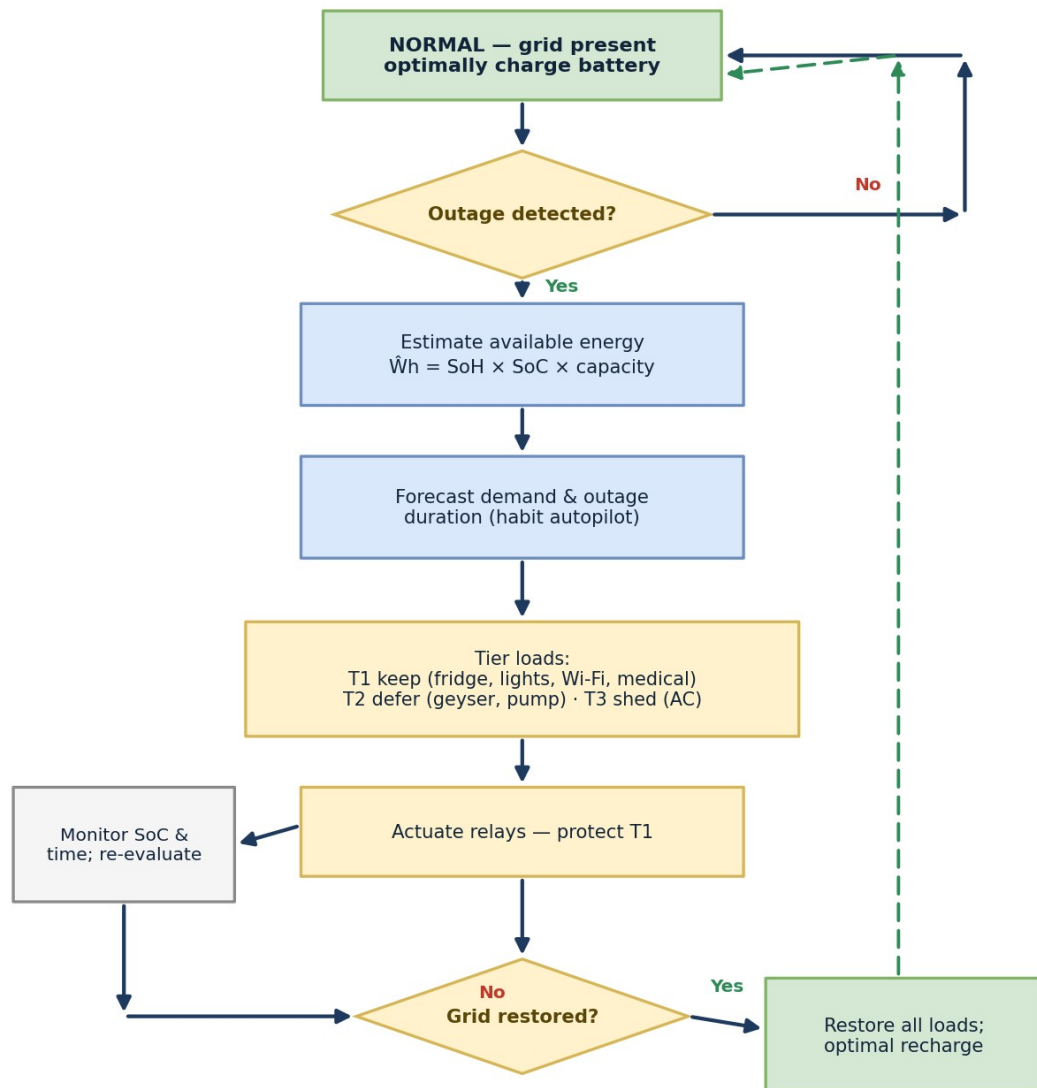


Exhibit 8. Autonomous blackout load-prioritisation control logic: a rule-based tiering core informed by a learned demand and outage-duration forecaster.

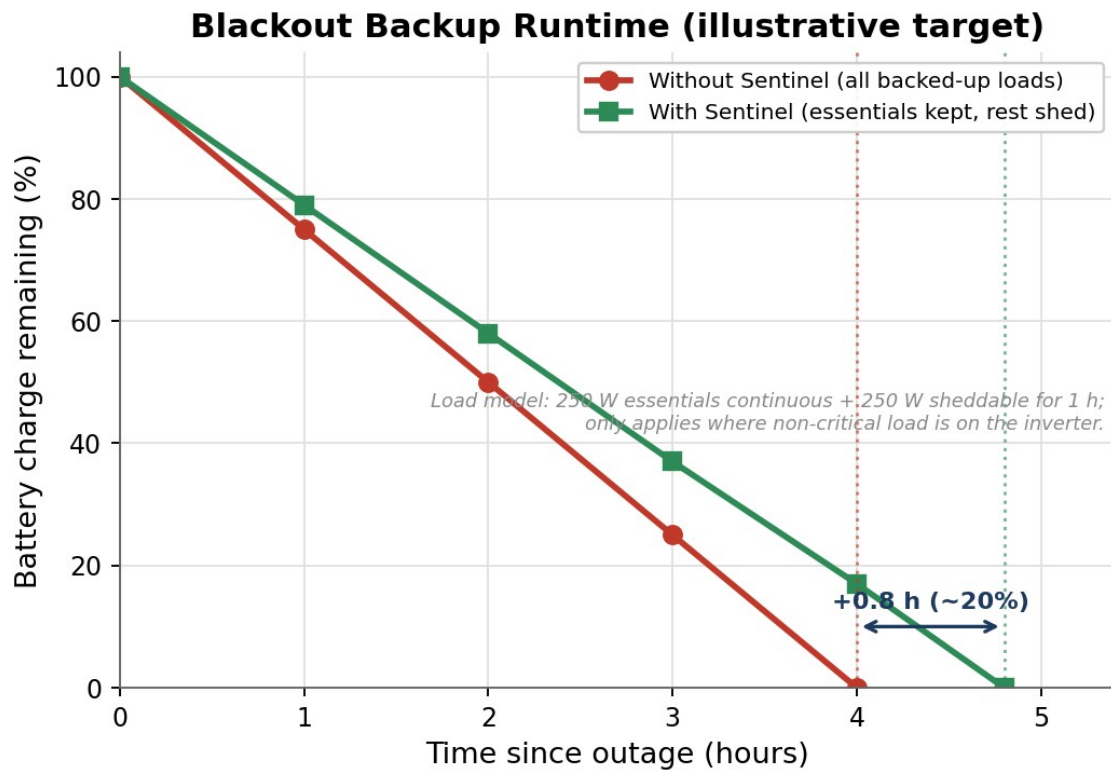


Exhibit 9. Blackout backup runtime (illustrative target, pre-validation): where non-critical loads are present on the inverter, protecting essentials and shedding the rest extends usable backup for essentials during an outage.

8. Engine 3: Energy Coach and Grid Shield

The Energy Coach uses non-intrusive load monitoring to disaggregate consumption from a low-frequency (≤ 1 Hz) current signal. We scope it honestly: low-frequency disaggregation reliably separates high-power loads such as air-conditioners, geysers, pumps and refrigerators at roughly 0.65 to 0.89 F-score on benchmark datasets [20], but low-power and overlapping loads and cross-home generalisation are known weaknesses, and the only public Indian dataset is a single Delhi home [21]. Sentinel therefore positions the Coach as directional insight with per-home calibration for headline loads, treating fine-grained disaggregation as a cloud-assisted roadmap item - valuable as time-of-use tariffs arrive with the national smart-meter rollout [29], and never over-claimed. The Grid Shield is scoped with equal care. Grid-origin voltage sags propagate almost instantly and have no locally observable precursor, so the honest capability is fast detection, not prediction: using a one-cycle time-domain RMS the Core detects a sag within about one cycle (a multi-cycle FFT handles harmonics). It cannot itself act faster than the inverter's own transfer switch, so its role is to log the event, alert the user and flag grid abuse for appliance-protection analytics - not an independent protective trip - targeting an IEC 61000-4-30 Class-S-like measurement rather than a certified Class-A instrument [18][19]. Only a self-caused local motor

start is genuinely anticipatable; seasonal and statistical risk is advisory only. Claiming detection rather than prediction is exactly the kind of precision expert judges reward.

9. Engine 4: Federated Fleet Intelligence

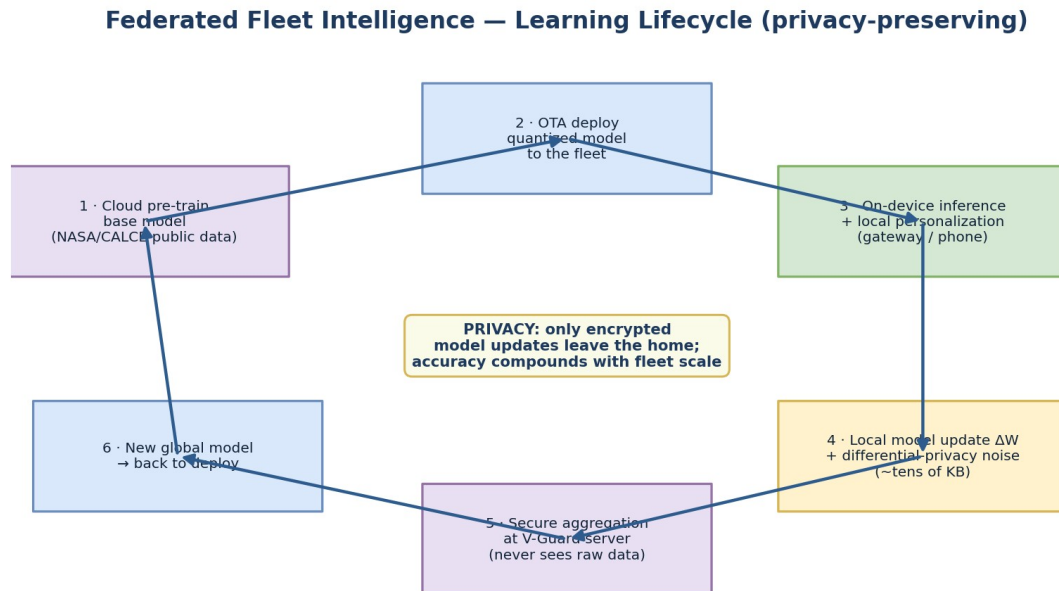


Exhibit 10. Federated learning lifecycle: cloud pre-training, OTA deployment, on-device inference with gateway/phone-tier local training, differentially-private updates, and secure server-side aggregation.

The moat is the data, not the mechanism. Because V-Guard owns the entire installed base and manufactures the cells, its real, un-copyable asset is proprietary aging and field data a competitor cannot buy; the primary learning path is therefore simple and robust - units log anonymised battery telemetry (voltage, current, temperature, throughput, depth-of-discharge histograms, none of it personally sensitive) and upload it opportunistically when power and connectivity allow, and V-Guard trains improved models centrally and pushes them over the air. On-device inference always runs offline; only learning needs the network. For connected units, an optional privacy-preserving federated-learning path adds edge personalisation without raw data leaving the home. Training deep models directly on the inverter microcontroller across a fleet is not realistic today; at most, sparse on-device fine-tuning of a fixed model is feasible, as demonstrated by on-device training under 256 kilobytes of memory [17]. The honest architecture is: the microcontroller runs quantised inference continuously; a home gateway or the owner's paired phone performs the actual local training round, because it has the memory, floating-point unit and reliable power; only model deltas of tens to hundreds of kilobytes are uploaded opportunistically when power and connectivity allow; and a V-Guard server aggregates them with FedAvg or DP-FTRL into an

improved global model that is pushed on the next update window. We use the Flower framework, which is built for exactly this intermittent, framework-agnostic deployment [14]. Privacy is handled properly: because model updates can leak training data through gradient inversion, Sentinel combines Secure Aggregation, so the server sees only the summed update [15], with Differential Privacy at a target budget of about ($\epsilon=4$, $\delta=1e-5$) per user-year via DP-FTRL, the production technique behind large-scale federated keyboards [16]. We are candid that this federated path depends on a capable second device and connectivity that much of the rural base lacks, and that privacy strengthens with cohort size - which is why it is an enhancement layered on the central path, not a precondition for the product. Either way, accuracy compounds as V-Guard's proprietary fleet data grows, something no competitor can rebuild without the same installed base (Exhibit 11).

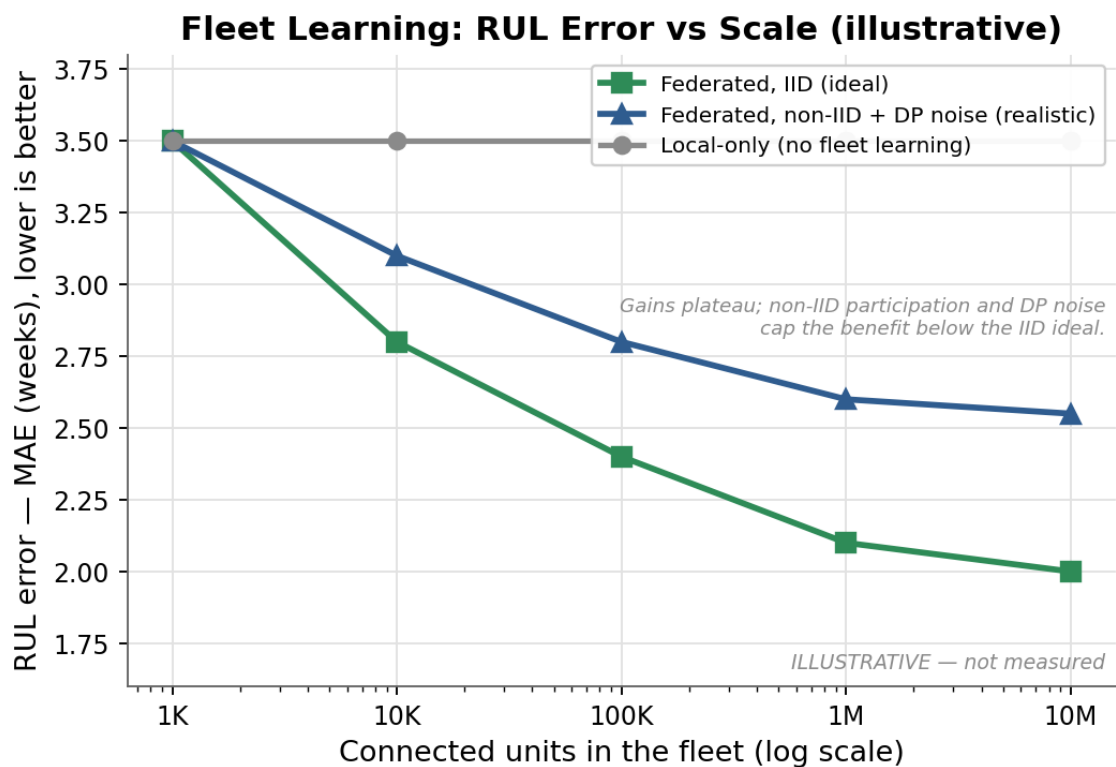


Exhibit 11. The data moat: model accuracy compounds as V-Guard's proprietary fleet data grows with the installed base - via central training, with optional federated learning - an advantage competitors cannot buy.

For that optional federated path the communication is light: updates are the size of the model (tens to hundreds of kilobytes), a device participates roughly once a day, and the protocol tolerates the intermittent participation a real Indian fleet exhibits - the same opportunistic, privacy-bounded, server-aggregated pattern already proven at national scale by production federated keyboards [16].

10. Key Parameters and Performance Targets

The design is specified by the parameters in Exhibit 14, spanning electrical sensing, computation, machine-learning thresholds, the federated protocol, product-level targets and the operating environment. These values define acceptance criteria for the prototype and the production module, and are the quantities the Phase-4 build will validate against.

11. Prototype and Validation Plan

Matching our team's machine-learning and full-stack strengths, the prototype is software-led. We will train a battery-RUL model on NASA and CALCE lithium-ion data augmented with synthetic Indian tubular duty cycles - demonstrating the method and the transfer-learning approach the in-house lead-acid aging campaign will later specialise - and report mean-absolute error, root-mean-square error and calibration of the predicted failure window on held-out cells. We will run a federated-learning simulation in Flower across at least five simulated homes, demonstrating that the shared model improves while no raw data leaves a client, with Secure Aggregation and Differential Privacy enabled. A dashboard application will visualise state-of-health, the RUL countdown, the autonomous load-prioritisation simulation and energy insights. As an optional hardware demonstration, the quantised RUL model will be flashed to an ESP32-S3 with a current-sensor breadboard to prove genuine on-device inference without any custom PCB or manufacturing. The CAD package in this submission (DXF, AutoLISP and script) visualises the physical module. Success is measured by RUL-window hit-rate, state-of-charge error within five percent, load-shed correctness, federated accuracy relative to a centralised baseline, and on-device latency and footprint. Prototype images and videos will be shared via a linked folder: [insert Google Drive link].

Each demonstration maps to a marked criterion: the RUL model to novelty and technology, the learning simulation to the privacy design, and the on-device demo with the bill of materials and CAD to feasibility. We will publish the methodology and held-out results so the numbers can be checked, not taken on trust.

12. Standards, Safety and Compliance

The mandatory Indian gate for the electronics is BIS certification under IS/IEC 62368-1, which is replacing IS 13252 Part 1 for information and communication technology equipment [22]; the host inverter remains under IEC 62040 for uninterruptible power systems [23], and the module must not degrade the host's compliance. Electromagnetic compatibility and immunity follow the IEC 61000 family, including electrostatic-discharge, radiated-immunity and surge tests, with power-quality measurement performed to an IEC 61000-4-30 Class-S-like level [18]. Any lithium cell on the module falls under IS 16046-2 and IEC 62133-2 [24]. RoHS and, for export, CE

marking under the Low-Voltage and EMC directives apply. Isolation on all mains-referenced sensing, creepage and clearance designed for 300 volts plus transients, fused sense taps, transient suppression on the high-voltage node, and fail-safe relay defaults are core safety requirements. Whether Sentinel ships as an embedded sub-assembly (under the host's IEC 62040) or a standalone accessory (its own IS/IEC 62368-1) sets the exact compliance path.

13. Feasibility and Manufacturing

Sentinel is a software-led upgrade to products V-Guard already builds at scale, so the path to revenue is short: new inverters embed the Core on the main board, and the installed base is addressed by the retrofit module. V-Guard's in-house battery manufacturing lets health models be co-designed with the cell chemistry - an advantage a third-party add-on cannot match - and the Kochi reliability lab provides the validation and aging-campaign infrastructure the data strategy requires. Because the compute, sensing and machine-learning building blocks are all commercially proven and individually cited above, the technical risk is integration and data, not invention.

Physically, the module is a signal-and-control unit that mounts on or beside the inverter and senses the battery through an external shunt or Hall sensor, so the 50-to-150-ampere battery current never crosses its PCB - the same split used by mass-produced battery monitors, and validated here against real V-Guard battery and inverter dimensions (Exhibit 12). A retrofit is a trained-installer job (shunt in the DC main, fused voltage tap, load contactors), not a consumer clip-on; V-Guard's service network is the natural install channel, and new-inverter integration avoids it entirely.

Installation Context — where the Sentinel AI Core sits and how it senses the battery

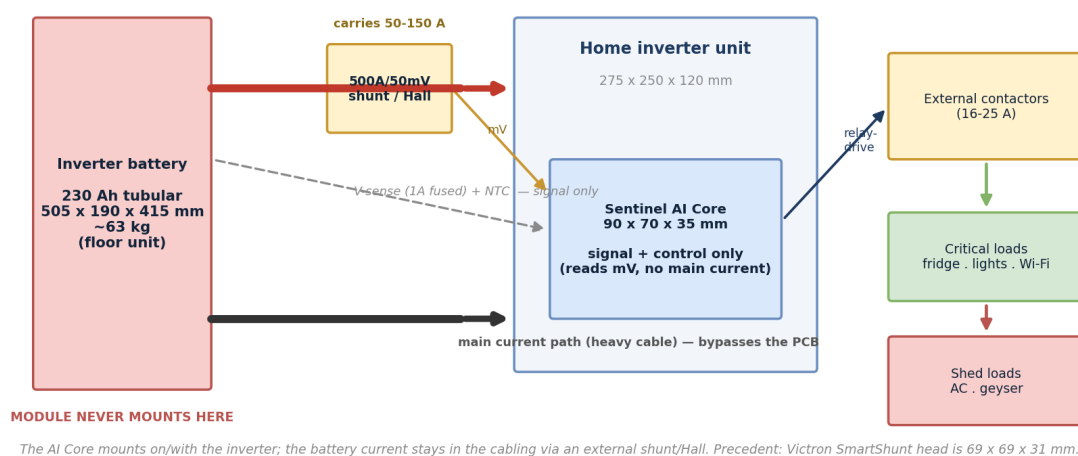


Exhibit 12. Installation context: the AI Core mounts with the inverter and senses the 505 x 190 x 415 mm, ~63 kg battery through an external 500A shunt or Hall sensor; only millivolt signals reach the PCB, and high-current loads are switched by external contactors.

Competitive Landscape — honest wins and gaps (with evidence)

Dimension	V-Guard Sentinel	Luminous	Havells	Sense (global)	Evidence / note
Predictive battery RUL (on-device ML)	✓✓	✗	✗	n/a	No Indian inverter ships ML end-of-life prediction
Autonomous offline load-prioritisation	✓✓	✗	✗	✗	Commercial load-shed is rule-based; none in Indian inverters
NILM energy disaggregation — maturity	~	✗	✗	✓✓	Sense WINS: production NILM + large labelled dataset
Rule-based smart charging shipping today	✓	✓	✓	n/a	Luminous Zello / Havells ship this now; Sentinel adds to it
Service & distribution footprint (India)	✓✓	✓✓	✓✓	✗	Havells/Luminous/V-Guard all strong; Sense absent in India
Fully offline / no-cloud operation	✓✓	✗	✗	✗	Sense is cloud-dependent; Sentinel core runs offline
Fleet data for model improvement	✓	✗	✗	✓✓	Sense has the richer NILM dataset today; V-Guard owns battery fleet
Mass-market India price point	✓	✓	✓	✗	Sense is premium; Indian brands compete on price

✓✓ strong · ✓ present · ~ partial/roadmap · ✗ absent · n/a not applicable. Competitor strengths (Sense NILM, Havells/Luminous footprint & shipping charging) shown honestly.

Exhibit 13. Competitive landscape (with evidence): Sentinel's genuine firsts shown alongside competitors' real strengths - Sense's NILM maturity, and Havells/Luminous' service footprint and shipping rule-based charging.

14. Business Value and Market Opportunity

India's home-UPS market is about 348 million dollars in 2024, growing to roughly 487 million by 2030; the inverter-battery market is on the order of 197 million dollars growing above six percent annually; stabilisers and geysers add further multi-hundred-million-dollar, mid-single-digit-growth markets [26][27]. V-Guard already sells across all of these through more than 100,000 retail touchpoints (distribution reach; the compatible installed base is a subset), so the addressable opportunity for a software-led upgrade is large, and the national smart-meter rollout with time-of-use tariffs creates demand for exactly the energy optimisation Sentinel provides [29]. The value compounds for V-Guard in four ways. First, warranty and service savings: V-Guard's warranty cost was about 1.52% of revenue, roughly 69 crore rupees in one recent year [25] - of which only the battery-attributable, preventable subset is addressable, a minority, since much warranty cost is manufacturing defect - and predictive maintenance converts a share of those emergency claims into planned interventions while defending against avoidable, warranty-voiding failures such as electrolyte neglect. Second, retention and a proprietary-data moat that deepens with every unit as fleet data grows - the hardest asset to copy. Third, well-timed aftermarket capture, stated honestly: a weeks-ahead warning does not by itself lock the replacement to V-Guard, since an informed owner could shop competitors, so the capture mechanism is convenience - a one-tap in-app reorder fulfilled by V-Guard's 100,000-touchpoint service network with doorstep swap and trade-in credit, making a genuine V-Guard battery the easiest choice rather than the cheapest generic.

Fourth, portfolio leverage, as the same compute platform extends to pumps, stabilisers and water heaters.

An illustrative sketch: a few-hundred-rupee incremental Core cost is recovered whenever it prevents one avoidable warranty replacement (a lead-acid battery costs several thousand rupees) or captures one well-timed aftermarket battery at V-Guard's margin - so only a small single-digit fraction of the ~69-crore-rupee warranty bill, or a modest reorder attach rate, makes the programme net-positive at fleet scale. We stop short of a full model, but the break-even logic is robust and does not require charging the consumer more.

15. Sustainability and Social Impact

Extending battery service life - temperature-compensated, anti-sulphation charging and reduced deep-discharge stress are established levers [6], targeting of the order of 15 to 30% pending field validation - and pinpointing replacement reduces premature disposal of lead-acid batteries - significant because an estimated 60 to 80% of India's used-lead-acid recycling is informal, and lead exposure already affects an estimated 275 million Indian children [28]. Optimised charging cuts wasted energy; predictive maintenance extends product lifespans across the portfolio and reduces material consumption. For price-sensitive and rural homes, resilient, self-managing backup protects food, study hours, connectivity and medical devices through the outages that remain a daily reality [1][2]. Sustainability here is not a bolt-on; it is a direct consequence of making the product predict, optimise and last.

16. Competitive Differentiation

As Exhibit 13 sets out, no mainstream Indian inverter today ships machine-learning battery-failure prediction, autonomous offline load-prioritisation, or fleet-data-driven model improvement; those three together are Sentinel's novelty, delivered offline at mass-market cost. Competitors lead on other axes - Sense on NILM maturity, Havells and Luminous on service footprint and shipping rule-based charging - which we credit openly.

17. Risks and Mitigations

The principal risk is over-claiming, which we mitigate by scoping every engine to the deployed-versus-research boundary and reporting windows and ranges rather than false precision. Model generalisation across chemistry, climate and home is addressed by our own aging campaign, federated refinement and per-home calibration with conservative confidence bounds. Privacy risk is met with Secure Aggregation and Differential Privacy and honestly modest early claims. Cost creep is controlled by the tiered bill of materials and reuse of existing sensing. Safety risk is met

with isolation, fail-safe relay defaults (loads default to energised on fault, with a watchdog), standards compliance and reliability-lab validation.

18. Roadmap and Vision

The rollout advances through validation gates, not calendar deadlines. Gate 1 (dataset acceptance): the in-house lead-acid aging campaign must reach a target RUL-window hit-rate on held-out cells before any RUL feature ships. Gate 2 (field pilot): battery-RUL and the autopilot run in a monitored Smart-Pro pilot fleet, validated against real replacements. Gate 3 (learning stability): the central-training pipeline, and the optional federated path, must show stable, non-degrading model updates before wide rollout. Only then does the platform extend across pumps, stabilisers and water heaters. The vision is that by 2030 every V-Guard product in a home runs on one shared AI platform, operating not as separate appliances but as a single self-aware system that predicts its own failures, manages the home's power, and keeps essentials running through any outage without the user thinking about it. Because every unit shipped feeds the same models with fresh fleet data, scale itself becomes an advantage rivals cannot rebuild, and V-Guard becomes the intelligence layer the Indian home runs on.

Exhibit 14. Key parameters and performance targets

Parameter	Target / range
Battery current sensing	EXTERNAL 500A/50mV (~0.1 mΩ) busbar shunt or ACS758 Hall (±200A); INA226/228 reads mV only — main current never on the PCB; auto-zero offset
Isolated voltage sensing	150–300 V via divider + reinforced isolation (AMC1311); thin sense leads FUSED 1A at terminal; ±5–10 mV (lead-acid)
Temperature	Battery-terminal NTC; optional ±0.1 °C digital (TMP117); operating 0–55 °C
Enclosure & mounting	~90 × 70 × 35 mm, PC/ABS UL94 V-0, IP20; mounts on/with the inverter (NOT the battery); high-current loads via external contactors
Compute / model	ESP32-S3 (Xtensa LX7, ESP-DSP/ESP-NN); int8 models 5–100 KB flash + ~8 KB activation RAM; 512 KB on-chip SRAM (RTOS/BLE/Wi-Fi headroom); inference <10 ms
SoC estimation	Coulomb count + OCV + EKF; error budget ±5%
SoH usable threshold	80% of rated capacity (end-of-life convention)
RUL output	State-of-health with uncertainty band; graded warning 'replace within N weeks'
Load tiers	T1 keep (fridge, lights, Wi-Fi, medical) · T2 defer (geyser, pump) · T3 shed (AC) — for loads actually

	on the inverter
Power-quality	1-cycle time-domain RMS (sag/swell events) + multi-cycle FFT (harmonics); 128–256 samples/cycle; IEC 61000-4-30 Class-S-like; detection not prediction
Learning path	Primary: opt-in anonymised fleet telemetry → central training. Optional: federated (gateway/phone) + Secure Aggregation + Differential Privacy (target $\epsilon \approx 4$, $\delta = 1e-5$ per user-year)
Product targets	Battery-life extension target ~15–30% (pending validation); backup extension ~10–20% where sheddable load exists; low active draw (radio duty-cycled, off during outage)
Compliance	IS/IEC 62368-1; IEC 62040 (host); IEC 61000 EMC; IS 16046-2 (if Li cell); RoHS/CE

Exhibit 15. Illustrative bill of materials (basic variant; premium detailed in hardware notes)

Item (basic variant)	Approx. INR
ESP32-S3 module (Wi-Fi + BLE)	250
INA226 monitor (reads battery-shunt mV)	60
NTC battery-terminal temperature	15
ULN2003 relay-driver (drives external contactors)	20
Wide-Vin buck (3.3 V) + LDO	45
UL94 V-0 enclosure	60
Connectors + 1 A-fused sense leads	50
Board subtotal (≈)	~500
External 500 A / 50 mV busbar shunt (reused if inverter has one)	300
BASIC standalone total (≈)	~800 (~500 if shunt reused)
Incremental over a metering inverter (shunt/relay/MCU shared)	~150–300

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